

Chapter 7 Exercises

SOLUTIONS TO RICK DURRETT'S
Probability: Theory and Examples

June 12, 2026

These solutions use the local Chapter 7 LLM/Viki knowledge set in `Probability/LLM_Viki_Dataset/Chapter7_Knowledge`. The cited knowledge IDs refer to entries in `chapter7_knowledge_pieces.jsonl`.

Exercise 7.1.1

Question. Given $s < t$, find

$$\mathbb{P}(B(s) > 0, B(t) > 0).$$

Hint: write the probability in terms of $B(s)$ and $B(t) - B(s)$.

Knowledge used. [durrett_ch7_brownian_definition](#);
[durrett_ch7_gaussian_process_characterization](#).

Solution. Write

$$X = B(s), \quad Y = B(t) - B(s).$$

Then X and Y are independent normal random variables with $X \sim N(0, s)$ and $Y \sim N(0, t - s)$. Let

$$U = \frac{X}{\sqrt{s}}, \quad V = \frac{Y}{\sqrt{t-s}}.$$

Then (U, V) has a rotationally symmetric standard normal distribution on $\mathbb{R}^2$, and the desired event is

$$\{U > 0, \sqrt{s}U + \sqrt{t-s}V > 0\}.$$

Because the joint density is radial, this probability is the angle of the corresponding sector divided by 2π .

The boundary lines are $U = 0$ and

$$\sqrt{s}U + \sqrt{t-s}V = 0, \quad \text{or} \quad V = -\sqrt{\frac{s}{t-s}}U.$$

The sector satisfying both inequalities has angle

$$\frac{\pi}{2} + \arctan \sqrt{\frac{s}{t-s}} = \frac{\pi}{2} + \arcsin \sqrt{\frac{s}{t}}.$$

Therefore

$$\mathbb{P}(B(s) > 0, B(t) > 0) = \frac{1}{4} + \frac{1}{2\pi} \arcsin \sqrt{\frac{s}{t}}.$$

Exercise 7.1.2

Question. Find

$$\mathbb{E}(B_1^2 B_2 B_3).$$

Knowledge used. [durrett_ch7_brownian_definition](#);
[durrett_ch7_gaussian_process_characterization](#).

Solution. Let

$$X = B_1, \quad Y = B_2 - B_1, \quad Z = B_3 - B_2.$$

By independent increments, X, Y, Z are independent $N(0, 1)$ random variables. Then

$$B_2 = X + Y, \quad B_3 = X + Y + Z,$$

so

$$\mathbb{E}(B_1^2 B_2 B_3) = \mathbb{E}[X^2(X + Y)(X + Y + Z)].$$

The terms involving the independent centered variable Z have mean zero, hence

$$\mathbb{E}(B_1^2 B_2 B_3) = \mathbb{E}[X^2(X + Y)^2].$$

Expanding and using independence and centering,

$$\mathbb{E}[X^2(X + Y)^2] = \mathbb{E}X^4 + 2\mathbb{E}X^3\mathbb{E}Y + \mathbb{E}X^2\mathbb{E}Y^2 = 3 + 0 + 1 = 4.$$

Thus

$$\boxed{\mathbb{E}(B_1^2 B_2 B_3) = 4}.$$

Exercise 7.1.3

Question. Let

$$W = \int_0^t B_s ds.$$

Find $\mathbb{E}W$ and $\mathbb{E}W^2$. What is the distribution of W ?

Knowledge used. [durrett_ch7_brownian_definition](#);
[durrett_ch7_gaussian_process_characterization](#).

Solution. Since $\mathbb{E}B_s = 0$ for all s ,

$$\mathbb{E}W = \int_0^t \mathbb{E}B_s ds = 0.$$

Also, using $\mathbb{E}(B_r B_s) = r \wedge s$,

$$\mathbb{E}W^2 = \mathbb{E}\left(\int_0^t B_r dr \int_0^t B_s ds\right) = \int_0^t \int_0^t (r \wedge s) dr ds.$$

By symmetry,

$$\int_0^t \int_0^t (r \wedge s) dr ds = 2 \int_0^t \int_0^s r dr ds = 2 \int_0^t \frac{s^2}{2} ds = \frac{t^3}{3}.$$

Hence

$$\boxed{\mathbb{E}W = 0, \quad \mathbb{E}W^2 = \frac{t^3}{3}}.$$

Finally, W is a limit in L^2 of Riemann sums

$$\sum_i B_{s_i} \Delta s_i,$$

and each such sum is a centered Gaussian random variable because Brownian motion is a Gaussian process. Therefore its L^2 limit W is also Gaussian. Thus

$$W \sim N\left(0, \frac{t^3}{3}\right).$$

Exercise 7.1.4

Question. Show that $A \in \mathcal{F}^o$ if and only if there is a sequence of times $t_1, t_2, \dots$ in $[0, \infty)$ and a Borel set $B \in \mathcal{B}(\mathbb{R}^{\{1,2,\dots\}})$ such that

$$A = \{\omega : (\omega(t_1), \omega(t_2), \dots) \in B\}.$$

In words, all events in $\mathcal{F}^o$ depend on only countably many coordinates.

Knowledge used. [durrett_ch7_kolmogorov_extension_construction](#); [durrett_ch7_brownian_definition](#).

Solution. Let Ω^o be the space of all real-valued functions on $[0, \infty)$, and let $\mathcal{F}^o$ be the σ -field generated by the coordinate maps $\omega \mapsto \omega(t)$.

First suppose that

$$A = \{\omega : (\omega(t_1), \omega(t_2), \dots) \in B\}$$

for some sequence $t_1, t_2, \dots$ and some $B \in \mathcal{B}(\mathbb{R}^{\{1,2,\dots\}})$. The map

$$\omega \mapsto (\omega(t_1), \omega(t_2), \dots)$$

is $\mathcal{F}^o/\mathcal{B}(\mathbb{R}^{\{1,2,\dots\}})$ -measurable, since each coordinate is $\mathcal{F}^o$ -measurable. Therefore $A \in \mathcal{F}^o$.

Conversely, let $\mathcal{G}$ be the collection of all subsets of Ω^o that depend on only countably many coordinates in the above sense. We claim that $\mathcal{G}$ is a σ -field. Complements are immediate: if A is represented using (t_n) and B , then A^c is represented using the same coordinates and B^c .

Now let $A_1, A_2, \dots \in \mathcal{G}$. For each j , choose coordinates $t_{j,1}, t_{j,2}, \dots$ and a Borel set B_j representing A_j . The union of the countable coordinate sets

$$\{t_{j,k} : j, k \geq 1\}$$

is still countable, so enumerate it as $u_1, u_2, \dots$. Each event A_j can be rewritten as a Borel condition on $(\omega(u_1), \omega(u_2), \dots)$. Hence $\bigcup_j A_j$ is also a Borel condition on the same countable list of coordinates. Thus $\mathcal{G}$ is a σ -field.

Every finite-dimensional cylinder event belongs to $\mathcal{G}$, since it depends on finitely many, hence countably many, coordinates. Because $\mathcal{F}^o$ is generated by the finite-dimensional cylinder events, we get

$$\mathcal{F}^o \subseteq \mathcal{G}.$$

The first paragraph proved $\mathcal{G} \subseteq \mathcal{F}^o$, so $\mathcal{G} = \mathcal{F}^o$. This proves the claim.

Exercise 7.1.5

Question. Looking at the proof of Theorem 7.1.6 carefully shows that if $\gamma > 5/6$, then B_t is not Holder continuous with exponent γ at any point in $[0, 1]$. Show, by considering k increments instead of 3, that the last conclusion is true for all

$$\gamma > \frac{1}{2} + \frac{1}{k}.$$

Knowledge used. [durrett_ch7_brownian_nowhere_differentiable](#);
[durrett_ch7_brownian_holder_half](#); [durrett_ch7_brownian_definition](#).

Solution. Fix $C < \infty$ and an integer $k \geq 3$. For $n \geq k$, define the event $A_n(C)$ that there is an $s \in [0, 1]$ such that

$$|B_t - B_s| \leq C|t - s|^\gamma \quad \text{whenever } |t - s| \leq k/n.$$

If Brownian motion is Holder continuous of exponent γ at some point with local constant C , then $A_n(C)$ occurs for all large n .

Now divide $[0, 1]$ into intervals of length $1/n$. If $A_n(C)$ occurs, then for some block of k consecutive increments near s , each increment must be small. More precisely, for a constant C_k depending only on C and k , the event $A_n(C)$ is contained in

$$\bigcup_{j=0}^{n-k} \bigcap_{\ell=1}^k \{|B_{(j+\ell)/n} - B_{(j+\ell-1)/n}| \leq C_k n^{-\gamma}\}.$$

The k increments in a fixed block are independent $N(0, 1/n)$ random variables. If $Z \sim N(0, 1/n)$, then

$$\mathbb{P}(|Z| \leq C_k n^{-\gamma}) = \mathbb{P}(|N(0, 1)| \leq C_k n^{1/2-\gamma}) \leq C'_k n^{1/2-\gamma}$$

for another constant C'_k . Therefore

$$\mathbb{P}(A_n(C)) \leq n \left(C'_k n^{1/2-\gamma} \right)^k = (C'_k)^k n^{1+k(1/2-\gamma)}.$$

If

$$\gamma > \frac{1}{2} + \frac{1}{k},$$

then $1 + k(1/2 - \gamma) < 0$, so $\mathbb{P}(A_n(C)) \rightarrow 0$.

Thus, for each fixed C , the probability that such a Holder point with constant C exists is 0. Taking the countable union over $C = 1, 2, \dots$ gives probability 0 for the existence of any Holder point of exponent γ in $[0, 1]$. Hence, almost surely, Brownian motion is not Holder continuous of exponent γ at any point of $[0, 1]$ whenever $\gamma > 1/2 + 1/k$.

Exercise 7.1.6

Question. Fix t and let

$$\Delta_{m,n} = B(tm2^{-n}) - B(t(m-1)2^{-n}).$$

Compute

$$\mathbb{E} \left(\sum_{m \leq 2^n} \Delta_{m,n}^2 - t \right)^2$$

and use Borel–Cantelli to conclude that

$$\sum_{m \leq 2^n} \Delta_{m,n}^2 \rightarrow t \quad \text{a.s. as } n \rightarrow \infty.$$

Knowledge used. [durrett_ch7_quadratic_variation_hint](#); [durrett_ch7_brownian_definition](#).

Solution. For fixed n , the increments

$$\Delta_{1,n}, \dots, \Delta_{2^n,n}$$

are independent and each has distribution $N(0, t2^{-n})$. Hence

$$\mathbb{E}\Delta_{m,n}^2 = t2^{-n}, \quad \mathbb{E}\Delta_{m,n}^4 = 3(t2^{-n})^2.$$

Therefore

$$\text{Var}(\Delta_{m,n}^2) = \mathbb{E}\Delta_{m,n}^4 - (\mathbb{E}\Delta_{m,n}^2)^2 = 2t^22^{-2n}.$$

Since the squared increments are independent,

$$\mathbb{E}\left(\sum_{m \leq 2^n} \Delta_{m,n}^2\right) = 2^n t 2^{-n} = t,$$

and

$$\mathbb{E}\left(\sum_{m \leq 2^n} \Delta_{m,n}^2 - t\right)^2 = \text{Var}\left(\sum_{m \leq 2^n} \Delta_{m,n}^2\right) = 2^n \cdot 2t^22^{-2n} = \boxed{2t^22^{-n}}.$$

Now let

$$Q_n = \sum_{m \leq 2^n} \Delta_{m,n}^2.$$

For any $\varepsilon > 0$, Chebyshev's inequality gives

$$\mathbb{P}(|Q_n - t| > \varepsilon) \leq \frac{\mathbb{E}(Q_n - t)^2}{\varepsilon^2} = \frac{2t^2}{\varepsilon^2} 2^{-n}.$$

Since

$$\sum_{n=1}^{\infty} \mathbb{P}(|Q_n - t| > \varepsilon) < \infty,$$

the Borel–Cantelli lemma implies that $|Q_n - t| > \varepsilon$ occurs only finitely often, almost surely. Taking ε over the positive rationals gives

$$\boxed{\sum_{m \leq 2^n} \Delta_{m,n}^2 \rightarrow t \quad \text{almost surely.}}$$

Exercise 7.2.1

Question. Let

$$T_0 = \inf\{s > 0 : B_s = 0\} \quad \text{and} \quad R = \inf\{t > 1 : B_t = 0\}.$$

R is for right or return. Use the Markov property at time 1 to get

$$\mathbb{P}_x(R > 1 + t) = \int p_1(x, y) \mathbb{P}_y(T_0 > t) dy.$$

Knowledge used. [durrett_ch7_markov_property](#);
[durrett_ch7_immediate_positive_negative_hits](#); [durrett_ch7_one_dimensional_recurrence](#).

Solution. The event $\{R > 1 + t\}$ says that, after time 1, Brownian motion does not hit 0 during the next time interval of length t . If we condition on $B_1 = y$, then the Markov property says that the post-1 process has the law of Brownian motion started at y . Therefore

$$\mathbb{P}_x(R > 1 + t \mid B_1) = \mathbb{P}_{B_1}(T_0 > t).$$

Taking expectations gives

$$\mathbb{P}_x(R > 1 + t) = \mathbb{E}_x \mathbb{P}_{B_1}(T_0 > t).$$

Since B_1 has density $p_1(x, y)$ under $\mathbb{P}_x$, this becomes

$$\boxed{\mathbb{P}_x(R > 1 + t) = \int_{\mathbb{R}} p_1(x, y) \mathbb{P}_y(T_0 > t) dy.}$$

When $y = 0$, $\mathbb{P}_0(T_0 > t) = 0$ for $t > 0$ by immediate return, which is consistent with the displayed formula.

Exercise 7.2.2

Question. Let

$$T_0 = \inf\{s > 0 : B_s = 0\} \quad \text{and} \quad L = \sup\{t \leq 1 : B_t = 0\}.$$

L is for left or last. Use the Markov property at time $0 < t < 1$ to conclude

$$\mathbb{P}_0(L \leq t) = \int p_t(0, y) \mathbb{P}_y(T_0 > 1 - t) dy.$$

Knowledge used. [durrett_ch7_markov_property](#);
[durrett_ch7_immediate_positive_negative_hits](#).

Solution. The event $\{L \leq t\}$ means that there is no zero of the Brownian path in the time interval $(t, 1]$. Conditional on $B_t = y$, the future process

$$B_{t+u}, \quad 0 \leq u \leq 1 - t,$$

has, by the Markov property, the law of Brownian motion started from y . Thus

$$\mathbb{P}_0(L \leq t \mid B_t) = \mathbb{P}_{B_t}(T_0 > 1 - t).$$

Taking expectations and using the transition density of B_t under $\mathbb{P}_0$ gives

$$\boxed{\mathbb{P}_0(L \leq t) = \int_{\mathbb{R}} p_t(0, y) \mathbb{P}_y(T_0 > 1 - t) dy.}$$

Again the value at $y = 0$ causes no difficulty, since $\mathbb{P}_0(T_0 > 1 - t) = 0$.

Exercise 7.2.3

Question. If $a < b$, then with probability one a is the limit of local maxima of B_t in (a, b) . So the set of local maxima of B_t is almost surely dense. However, unlike the zero set, it is countable.

Knowledge used. [durrett_ch7_markov_property](#); [durrett_ch7_blumenthal_zero_one](#); [durrett_ch7_immediate_positive_negative_hits](#); [durrett_ch7_brownian_definition](#).

Solution. Fix $a < b$. Choose deterministic intervals

$$[u_n, v_n] \subset (a, b), \quad u_n \downarrow a, \quad v_n \downarrow a, \quad u_n < v_n.$$

For instance, for large n one may take $u_n = a + (b-a)/(n+1)$ and $v_n = a + (b-a)/n$. By continuity, Brownian motion attains its maximum on each compact interval $[u_n, v_n]$; let τ_n be a time at which this maximum is attained.

We claim that, with probability one, τ_n is not an endpoint. At the left endpoint u_n , the increment process

$$B_{u_n+s} - B_{u_n}, \quad s \geq 0,$$

is Brownian motion started from 0. By immediate crossing, it is positive at arbitrarily small positive times almost surely. Hence u_n cannot be the maximum point of $[u_n, v_n]$.

At the right endpoint v_n , consider the reversed increment process

$$X_s = B_{v_n-s} - B_{v_n}, \quad 0 \leq s \leq v_n - u_n.$$

Its finite-dimensional distributions are those of Brownian motion started from 0 on this finite interval. Therefore it is also positive at arbitrarily small positive times almost surely. Equivalently, there are times just to the left of v_n at which B is larger than B_{v_n} . Thus v_n cannot be the maximum point of $[u_n, v_n]$.

Therefore $\tau_n \in (u_n, v_n)$ almost surely. Since B_{τ_n} is the maximum on the whole interval $[u_n, v_n]$, the time τ_n is a local maximum of the Brownian path. Also $\tau_n \rightarrow a$, because $u_n, v_n \rightarrow a$. This proves that a is almost surely a limit of local maxima lying in (a, b) .

Applying the result to all rational intervals shows that local maxima are almost surely dense in $[0, \infty)$. The countability comment is consistent with this: for Brownian paths, local maxima are almost surely strict, and strict local maxima of a continuous function form a countable set by assigning to each one a rational interval on which it is the unique maximum.

Exercise 7.2.4

Question.

- (i) Suppose $f(t) > 0$ for all $t > 0$. Use Theorem 7.2.3 to conclude that

$$\limsup_{t \downarrow 0} \frac{B(t)}{f(t)} = c \quad \mathbb{P}_0\text{-a.s.},$$

where $c \in [0, \infty]$ is a constant.

- (ii) Show that if $f(t) = \sqrt{t}$, then $c = \infty$, so with probability one Brownian paths are not Hölder continuous of order $1/2$ at 0.

Knowledge used. [durrett_ch7_blumenthal_zero_one](#);
[durrett_ch7_immediate_positive_negative_hits](#); [durrett_ch7_brownian_scaling](#);
[durrett_ch7_brownian_holder_half](#).

Solution. Let

$$L = \limsup_{t \downarrow 0} \frac{B(t)}{f(t)}.$$

For each real r , the event $\{L \leq r\}$ depends only on the values of Brownian motion in arbitrarily small intervals after time 0. Hence

$$\{L \leq r\} \in \mathcal{F}_0^+.$$

By Blumenthal's 0-1 law,

$$\mathbb{P}_0(L \leq r) \in \{0, 1\} \quad \text{for every } r.$$

Since Brownian motion started at 0 immediately visits positive values, $L \geq 0$ almost surely. Define

$$c = \inf\{r \in \mathbb{Q} : r \geq 0, \mathbb{P}_0(L \leq r) = 1\},$$

with the convention $c = \infty$ if the set is empty. The preceding 0-1 statement implies that $L = c$ almost surely. This proves (i).

Now take $f(t) = \sqrt{t}$. Fix $M > 0$ and $\delta > 0$. Since $B_\delta/\sqrt{\delta} \sim N(0, 1)$,

$$\mathbb{P}_0 \left(\sup_{0 < u \leq \delta} \frac{B_u}{\sqrt{u}} > M \right) \geq \mathbb{P}_0 \left(\frac{B_\delta}{\sqrt{\delta}} > M \right) = \mathbb{P}(N(0, 1) > M) > 0.$$

Letting $\delta \downarrow 0$ through a decreasing sequence gives

$$\mathbb{P}_0 \left(\limsup_{t \downarrow 0} \frac{B_t}{\sqrt{t}} \geq M \right) > 0.$$

This event lies in $\mathcal{F}_0^+$, so Blumenthal's 0-1 law upgrades its probability to 1. Since M is arbitrary,

$$\limsup_{t \downarrow 0} \frac{B_t}{\sqrt{t}} = \infty \quad \mathbb{P}_0\text{-a.s.}$$

Thus $c = \infty$.

If the Brownian path were Hölder continuous of order $1/2$ at 0, then there would be constants $C, \eta > 0$ such that

$$|B_t - B_0| \leq C\sqrt{t} \quad \text{for } 0 < t < \eta.$$

Since $B_0 = 0$, this would make $B_t/\sqrt{t}$ bounded above near 0, contradicting the infinite limsup just proved. Therefore Brownian paths are almost surely not Hölder continuous of order $1/2$ at 0.

Exercise 7.3.1

Question. Let A be an F_σ , that is, a countable union of closed sets. Show that

$$T_A = \inf\{t : B_t \in A\}$$

is a stopping time.

Knowledge used. [durrett_ch7_stopping_time_definition](#);
[durrett_ch7_hitting_times_open_closed](#); [durrett_ch7_stopping_time_limits](#).

Solution. Write

$$A = \bigcup_{n=1}^{\infty} K_n,$$

where each K_n is closed. Let

$$T_n = \inf\{t : B_t \in K_n\}.$$

By the closed-set hitting-time theorem, each T_n is a stopping time. Moreover,

$$T_A = \inf_{n \geq 1} T_n,$$

because hitting the union is the same as hitting at least one of the closed sets.

For every $t \geq 0$,

$$\{T_A < t\} = \left\{ \inf_n T_n < t \right\} = \bigcup_{n=1}^{\infty} \{T_n < t\}.$$

Each event $\{T_n < t\}$ is in $\mathcal{F}_t$, and $\mathcal{F}_t$ is closed under countable unions. Hence $\{T_A < t\} \in \mathcal{F}_t$ for all t , so T_A is a stopping time.

Exercise 7.3.2

Question. If S and T are stopping times, then

$$S \wedge T = \min\{S, T\}, \quad S \vee T = \max\{S, T\}, \quad S + T$$

are also stopping times. In particular, if $t \geq 0$, then $S \wedge t$, $S \vee t$, and $S + t$ are stopping times.

Knowledge used. [durrett_ch7_stopping_time_definition](#); [durrett_ch7_stopping_time_limits](#).

Solution. For the minimum,

$$\{S \wedge T < t\} = \{S < t\} \cup \{T < t\} \in \mathcal{F}_t.$$

Thus $S \wedge T$ is a stopping time.

For the maximum, use the equivalent criterion $\{U \leq t\} \in \mathcal{F}_t$ for stopping times under a right-continuous filtration. Then

$$\{S \vee T \leq t\} = \{S \leq t\} \cap \{T \leq t\} \in \mathcal{F}_t.$$

So $S \vee T$ is a stopping time.

For the sum, for $t > 0$,

$$\{S + T < t\} = \bigcup_{\substack{q \in \mathbb{Q} \\ 0 < q < t}} \{S < q\} \cap \{T < t - q\}.$$

Indeed, if $S + T < t$, choose a rational q with $S < q < t - T$. Conversely, any such q implies $S + T < t$. The event $\{S < q\}$ lies in $\mathcal{F}_q \subseteq \mathcal{F}_t$, and $\{T < t - q\}$ lies in $\mathcal{F}_{t-q} \subseteq \mathcal{F}_t$, so the countable union is in $\mathcal{F}_t$. Therefore $S + T$ is a stopping time.

Taking T to be the deterministic stopping time identically equal to t gives the conclusions for $S \wedge t$, $S \vee t$, and $S + t$.

Exercise 7.3.3

Question. Let T_n be a sequence of stopping times. Show that

$$\sup_n T_n, \quad \inf_n T_n, \quad \limsup_n T_n, \quad \liminf_n T_n$$

are stopping times.

Knowledge used. [durrett_ch7_stopping_time_definition](#); [durrett_ch7_stopping_time_limits](#).

Solution. For the infimum,

$$\left\{ \inf_n T_n < t \right\} = \bigcup_{n=1}^{\infty} \{T_n < t\} \in \mathcal{F}_t,$$

so $\inf_n T_n$ is a stopping time.

For the supremum, use the equivalent $\leq$ formulation:

$$\left\{ \sup_n T_n \leq t \right\} = \bigcap_{n=1}^{\infty} \{T_n \leq t\} \in \mathcal{F}_t.$$

Thus $\sup_n T_n$ is a stopping time.

Now write

$$\liminf_{n \rightarrow \infty} T_n = \sup_{m \geq 1} \inf_{n \geq m} T_n, \quad \limsup_{n \rightarrow \infty} T_n = \inf_{m \geq 1} \sup_{n \geq m} T_n.$$

The inner infima and suprema are stopping times by the first two parts, and applying the same two closure properties once more shows that $\liminf_n T_n$ and $\limsup_n T_n$ are stopping times.

Exercise 7.3.4

Question. Let S be a stopping time, let $A \in \mathcal{F}_S$, and let

$$R = \begin{cases} S, & \text{on } A, \\ \infty, & \text{on } A^c. \end{cases}$$

Show that R is a stopping time.

Knowledge used. [durrett_ch7_stopping_time_definition](#); [durrett_ch7_random_shift_and_F_S](#).

Solution. For each $t \geq 0$,

$$\{R \leq t\} = A \cap \{S \leq t\}.$$

Since $A \in \mathcal{F}_S$, the defining property of $\mathcal{F}_S$ gives

$$A \cap \{S \leq t\} \in \mathcal{F}_t.$$

Therefore $\{R \leq t\} \in \mathcal{F}_t$ for every t , and hence R is a stopping time.

Exercise 7.3.5

Question. Let S and T be stopping times. Show that

$$\{S < T\}, \quad \{S > T\}, \quad \{S = T\}$$

are in $\mathcal{F}_S$ and in $\mathcal{F}_T$.

Knowledge used. [durrett_ch7_stopping_time_definition](#); [durrett_ch7_random_shift_and_F_S](#).

Solution. We use the equivalent characterization of $\mathcal{F}_S$ with $A \cap \{S < t\} \in \mathcal{F}_t$ for every t .

First,

$$\{S < T\} \cap \{S < t\} = \bigcup_{\substack{q \in \mathbb{Q} \\ q < t}} \{S < q\} \cap \{q < T\}.$$

The event $\{S < q\}$ is in $\mathcal{F}_q$, and $\{q < T\} = \{T \leq q\}^c$ is also in $\mathcal{F}_q$. Since $q < t$, each term lies in $\mathcal{F}_t$, so the countable union lies in $\mathcal{F}_t$. Hence $\{S < T\} \in \mathcal{F}_S$.

Next,

$$\{S > T\} \cap \{S < t\} = \bigcup_{\substack{q \in \mathbb{Q} \\ q < t}} \{T < q\} \cap \{q < S\} \cap \{S < t\}.$$

Each event in the intersection is measurable by time t , so $\{S > T\} \in \mathcal{F}_S$. Finally,

$$\{S = T\} = \{S < T\}^c \cap \{S > T\}^c,$$

so $\{S = T\} \in \mathcal{F}_S$ as well.

The same argument with the roles of S and T interchanged shows that all three events also belong to $\mathcal{F}_T$.

Exercise 7.3.6

Question. Let S and T be stopping times. Show that

$$\mathcal{F}_S \cap \mathcal{F}_T = \mathcal{F}_{S \wedge T}.$$

Knowledge used. [durrett_ch7_stopping_time_definition](#); [durrett_ch7_random_shift_and_F_S](#); [durrett_ch7_stopping_time_limits](#).

Solution. Since $S \wedge T \leq S$ and $S \wedge T \leq T$, information available at time $S \wedge T$ is available at both later stopping times. Thus

$$\mathcal{F}_{S \wedge T} \subseteq \mathcal{F}_S \cap \mathcal{F}_T.$$

For the reverse inclusion, let $A \in \mathcal{F}_S \cap \mathcal{F}_T$. We must show that $A \in \mathcal{F}_{S \wedge T}$. For every $t \geq 0$,

$$A \cap \{S \wedge T \leq t\} = A \cap (\{S \leq t\} \cup \{T \leq t\}).$$

Hence

$$A \cap \{S \wedge T \leq t\} = (A \cap \{S \leq t\}) \cup (A \cap \{T \leq t\}).$$

Because $A \in \mathcal{F}_S$, the first term is in $\mathcal{F}_t$; because $A \in \mathcal{F}_T$, the second term is in $\mathcal{F}_t$. Therefore their union is in $\mathcal{F}_t$. This proves $A \in \mathcal{F}_{S \wedge T}$.

Combining the two inclusions gives

$$\boxed{\mathcal{F}_S \cap \mathcal{F}_T = \mathcal{F}_{S \wedge T}}.$$

Exercise 7.4.1

Question. Let $B_t = (B_t^1, B_t^2)$ be a two-dimensional Brownian motion starting from 0. Let

$$T_a = \inf\{t : B_t^2 = a\}.$$

Show that $B^1(T_a)$, $a \geq 0$, has independent increments and

$$B^1(T_a) \stackrel{d}{=} aB^1(T_1).$$

Use this to conclude that $B^1(T_a)$ has a Cauchy distribution.

Knowledge used. [durrett_ch7_d_dimensional_brownian](#);
[durrett_ch7_hitting_time_subordinator](#); [durrett_ch7_hitting_time_scaling](#);
[durrett_ch7_hitting_time_density](#).

Solution. The two coordinates B^1 and B^2 are independent one-dimensional Brownian motions. The random time T_a is determined only by the second coordinate.

Let

$$0 = a_0 < a_1 < \cdots < a_n.$$

Conditioning on the hitting times $T_{a_0}, T_{a_1}, \dots, T_{a_n}$, the increments

$$B^1(T_{a_j}) - B^1(T_{a_{j-1}}), \quad 1 \leq j \leq n,$$

are independent centered normal random variables with variances $T_{a_j} - T_{a_{j-1}}$. The process $\{T_a : a \geq 0\}$ has stationary independent increments as a function of the level a , so averaging over the hitting times preserves independence. Therefore

$$X_a := B^1(T_a), \quad a \geq 0,$$

has independent increments.

Next compute its characteristic function. Since B^1 is independent of T_a ,

$$\mathbb{E}e^{iuB^1(T_a)} = \mathbb{E}\left[\mathbb{E}\left(e^{iuB^1(T_a)} \mid T_a\right)\right] = \mathbb{E}e^{-u^2T_a/2}.$$

Brownian hitting-time scaling gives $T_a \stackrel{d}{=} a^2T_1$, hence

$$\mathbb{E}e^{iuB^1(T_a)} = \mathbb{E}e^{-(au)^2T_1/2} = \mathbb{E}e^{iu aB^1(T_1)}.$$

Thus

$$\boxed{B^1(T_a) \stackrel{d}{=} aB^1(T_1)}.$$

Finally, using the first-passage density

$$\mathbb{P}(T_a \in ds) = \frac{a}{\sqrt{2\pi s^3}} e^{-a^2/(2s)} ds, \quad a > 0,$$

or equivalently its Laplace transform, we get

$$\mathbb{E}e^{-u^2T_a/2} = e^{-a|u|}.$$

This is the characteristic function of the centered Cauchy distribution with scale parameter a , whose density is

$$\boxed{f_a(x) = \frac{1}{\pi} \frac{a}{a^2 + x^2}, \quad x \in \mathbb{R}, \ a > 0.}$$

For $a = 0$, $B^1(T_0) = 0$ is degenerate.

Exercise 7.4.2

Question. Use (7.2.3) to show that

$$R = \inf\{t > 1 : B_t = 0\}$$

has probability density

$$\mathbb{P}_0(R \in d(1+t))/dt = \frac{1}{\pi t^{1/2}(1+t)}.$$

Knowledge used. [durrett_ch7_markov_property](#); [durrett_ch7_hitting_time_density](#); [durrett_ch7_immediate_positive_negative_hits](#).

Solution. From Exercise 7.2.1 with $x = 0$,

$$\mathbb{P}_0(R > 1+t) = \int_{\mathbb{R}} p_1(0, y) \mathbb{P}_y(T_0 > t) dy.$$

For $y \neq 0$, the hitting time of 0 by Brownian motion started at y has the same law as the hitting time of $|y|$ by Brownian motion started at 0. Hence its density is

$$\frac{|y|}{\sqrt{2\pi r^3}} e^{-y^2/(2r)}, \quad r > 0.$$

Therefore the density of R at $1+t$ is obtained by differentiating the survival function:

$$f_R(1+t) = \int_{\mathbb{R}} p_1(0, y) \frac{|y|}{\sqrt{2\pi t^3}} e^{-y^2/(2t)} dy.$$

Substitute $p_1(0, y) = (2\pi)^{-1/2} e^{-y^2/2}$:

$$f_R(1+t) = \frac{1}{2\pi t^{3/2}} \int_{\mathbb{R}} |y| \exp\left\{-\frac{1+t}{2t} y^2\right\} dy.$$

Since

$$\int_{\mathbb{R}} |y| e^{-cy^2} dy = \frac{1}{c}, \quad c > 0,$$

with $c = (1+t)/(2t)$, we obtain

$$f_R(1+t) = \frac{1}{2\pi t^{3/2}} \cdot \frac{2t}{1+t} = \boxed{\frac{1}{\pi t^{1/2}(1+t)}}.$$

Exercise 7.4.3

Question.

- (a) Generalize the proof of (7.4.5) to conclude that if $u < v \leq a$, then

$$\mathbb{P}_0(T_a < t, u < B_t < v) = \mathbb{P}_0(2a - v < B_t < 2a - u).$$

- (b) Let (u, v) shrink down to x in the preceding identity to conclude that, for $x < a$,

$$\mathbb{P}_0(T_a < t, B_t \in dx) = p_t(0, 2a - x) dx.$$

- (c) Use $\{T_a < t\} = \{M_t > a\}$, where $M_t = \max_{0 \leq s \leq t} B_s$, and differentiate with respect to a to obtain the joint density

$$f_{(M_t, B_t)}(a, x) = \frac{2(2a - x)}{\sqrt{2\pi t^3}} e^{-(2a-x)^2/(2t)}.$$

Knowledge used. [durrett_ch7_reflection_principle](#); [durrett_ch7_strong_markov_property](#); [durrett_ch7_hitting_time_density](#).

Solution. For part (a), condition at the first hitting time T_a . On $\{T_a < t\}$, the post- T_a process

$$B_{T_a+r} - a, \quad r \geq 0,$$

is Brownian motion started from 0, independent of the past. By symmetry, after the first hit of a , paths ending below a in the interval (u, v) can be reflected across the horizontal line a and matched with paths whose endpoint lies in $(2a - v, 2a - u)$. Therefore

$$\boxed{\mathbb{P}_0(T_a < t, u < B_t < v) = \mathbb{P}_0(2a - v < B_t < 2a - u)}.$$

For part (b), divide by $v - u$ and let $u, v \rightarrow x < a$. Since B_t has density $p_t(0, \cdot)$,

$$\boxed{\mathbb{P}_0(T_a < t, B_t \in dx) = p_t(0, 2a - x) dx, \quad x < a.}$$

This is the subprobability density of B_t on the event that the path has crossed a before time t and ends below a .

For part (c), note that $\{T_a < t\} = \{M_t > a\}$. Thus for $x < a$,

$$\mathbb{P}_0(M_t > a, B_t \in dx) = p_t(0, 2a - x) dx.$$

The joint density of (M_t, B_t) is the negative derivative in a :

$$f_{(M_t, B_t)}(a, x) = -\frac{\partial}{\partial a} p_t(0, 2a - x).$$

Since

$$p_t(0, z) = \frac{1}{\sqrt{2\pi t}} e^{-z^2/(2t)},$$

we have

$$-\frac{\partial}{\partial a} p_t(0, 2a - x) = \frac{2(2a - x)}{t} p_t(0, 2a - x).$$

Therefore

$$\boxed{f_{(M_t, B_t)}(a, x) = \frac{2(2a - x)}{\sqrt{2\pi t^3}} e^{-(2a-x)^2/(2t)}},$$

for $a \geq 0$ and $x < a$.

Exercise 7.4.4

Question. Let $A_{s,t}$ be the event that Brownian motion has at least one zero in $[s, t]$. Show that

$$\mathbb{P}_0(A_{s,t}) = \frac{2}{\pi} \arccos \sqrt{\frac{s}{t}}.$$

Hint: To get started note that

$$\mathbb{P}_0(A_{s,t}) = 2 \int_0^\infty (2\pi s)^{-1/2} e^{-x^2/(2s)} \mathbb{P}_x(T_0 \leq t-s) dx$$

and use the formula for the density function of T_x . At the end, you may need some trigonometry to convert arctangent into arccosine.

Knowledge used. [durrett_ch7_markov_property](#); [durrett_ch7_hitting_time_density](#); [durrett_ch7_zero_set_structure](#).

Solution. Condition on $B_s = x$. By symmetry and the Markov property,

$$\mathbb{P}_0(A_{s,t}) = 2 \int_0^\infty (2\pi s)^{-1/2} e^{-x^2/(2s)} \mathbb{P}_x(T_0 \leq t-s) dx.$$

For $x > 0$, the density of T_0 under $\mathbb{P}_x$ is the same as the density of T_x under $\mathbb{P}_0$:

$$\mathbb{P}_x(T_0 \in dr) = \frac{x}{\sqrt{2\pi r^3}} e^{-x^2/(2r)} dr.$$

Thus

$$\mathbb{P}_0(A_{s,t}) = 2 \int_0^\infty (2\pi s)^{-1/2} e^{-x^2/(2s)} \int_0^{t-s} \frac{x}{\sqrt{2\pi r^3}} e^{-x^2/(2r)} dr dx.$$

Interchanging the order of integration gives

$$\mathbb{P}_0(A_{s,t}) = \frac{1}{\pi\sqrt{s}} \int_0^{t-s} r^{-3/2} \int_0^\infty x \exp\left\{-\frac{x^2}{2}\left(\frac{1}{s} + \frac{1}{r}\right)\right\} dx dr.$$

The inner integral is

$$\frac{sr}{s+r},$$

so

$$\mathbb{P}_0(A_{s,t}) = \frac{\sqrt{s}}{\pi} \int_0^{t-s} \frac{r^{-1/2}}{s+r} dr.$$

Let $u = \sqrt{r/s}$. Then $r = su^2$ and $dr = 2su du$, so

$$\frac{\sqrt{s}}{\pi} \int_0^{t-s} \frac{r^{-1/2}}{s+r} dr = \frac{2}{\pi} \int_0^{\sqrt{(t-s)/s}} \frac{du}{1+u^2}.$$

Therefore

$$\mathbb{P}_0(A_{s,t}) = \frac{2}{\pi} \arctan \sqrt{\frac{t-s}{s}}.$$

If $\theta = \arctan \sqrt{(t-s)/s}$, then

$$\cos \theta = \frac{1}{\sqrt{1 + (t-s)/s}} = \sqrt{\frac{s}{t}},$$

so $\theta = \arccos \sqrt{s/t}$. Hence

$$\boxed{\mathbb{P}_0(A_{s,t}) = \frac{2}{\pi} \arccos \sqrt{\frac{s}{t}}}.$$

Exercise 7.5.1

Question. Let

$$T = \inf\{t : B_t \notin (-a, a)\}.$$

Show that

$$\mathbb{E}_0 e^{-\lambda T} = \frac{1}{\cosh(a\sqrt{2\lambda})}.$$

Knowledge used. [durrett_ch7_optional_stopping_bounded](#);
[durrett_ch7_exponential_martingale](#); [durrett_ch7_expected_interval_exit_time](#).

Solution. Let

$$\alpha = \sqrt{2\lambda}.$$

Since both

$$e^{\alpha B_t - \alpha^2 t/2} \quad \text{and} \quad e^{-\alpha B_t - \alpha^2 t/2}$$

are martingales, their average

$$M_t = e^{-\lambda t} \cosh(\alpha B_t)$$

is also a martingale. Apply optional stopping to $T \wedge n$:

$$1 = M_0 = \mathbb{E}_0 \left[e^{-\lambda(T \wedge n)} \cosh(\alpha B_{T \wedge n}) \right].$$

Letting $n \rightarrow \infty$, the bounded convergence argument on the stopped exit event gives $B_T = \pm a$ and hence

$$1 = \mathbb{E}_0 \left[e^{-\lambda T} \cosh(\alpha B_T) \right] = \cosh(\alpha a) \mathbb{E}_0 e^{-\lambda T}.$$

Therefore

$$\boxed{\mathbb{E}_0 e^{-\lambda T} = \frac{1}{\cosh(a\sqrt{2\lambda})}}.$$

Exercise 7.5.2

Question. The point of this exercise is to get information about the amount of time it takes Brownian motion with drift $-b$,

$$X_t = B_t - bt,$$

to hit level a . Let

$$\tau = \inf\{t : B_t = a + bt\}, \quad a > 0.$$

Use the martingale $e^{\theta B_t - \theta^2 t/2}$ with

$$\theta = b + \sqrt{b^2 + 2\lambda}$$

to show

$$\mathbb{E}_0 e^{-\lambda \tau} = \exp\{-a(b + \sqrt{b^2 + 2\lambda})\}.$$

Letting $\lambda \rightarrow 0$ gives

$$\mathbb{P}_0(\tau < \infty) = e^{-2ab}.$$

Knowledge used. [durrett_ch7_exponential_martingale](#);
[durrett_ch7_optional_stopping_bounded](#).

Solution. Let

$$\theta = b + \sqrt{b^2 + 2\lambda}.$$

Then

$$\frac{\theta^2}{2} - b\theta = \lambda, \quad \text{or equivalently} \quad b\theta - \frac{\theta^2}{2} = -\lambda.$$

The exponential martingale

$$M_t = e^{\theta B_t - \theta^2 t/2}$$

stopped at $\tau \wedge n$ satisfies

$$1 = \mathbb{E}_0 M_{\tau \wedge n}.$$

On $\{\tau \leq n\}$, $B_\tau = a + b\tau$, so

$$M_\tau = \exp\{\theta a + (b\theta - \theta^2/2)\tau\} = e^{\theta a} e^{-\lambda \tau}.$$

On $\{\tau > n\}$, we have $B_n < a + bn$, and therefore

$$M_n \leq \exp\{\theta a + (b\theta - \theta^2/2)n\} = e^{\theta a} e^{-\lambda n}.$$

Letting $n \rightarrow \infty$ gives

$$1 = e^{\theta a} \mathbb{E}_0 e^{-\lambda \tau},$$

and hence

$$\boxed{\mathbb{E}_0 e^{-\lambda \tau} = e^{-a(b + \sqrt{b^2 + 2\lambda})}}.$$

For $b > 0$, letting $\lambda \downarrow 0$ gives

$$\boxed{\mathbb{P}_0(\tau < \infty) = e^{-2ab}}.$$

Exercise 7.5.3

Question. Let

$$\sigma = \inf\{t : B_t \notin (a, b)\}, \quad \lambda > 0.$$

Use the strong Markov property to show

$$\mathbb{E}_x e^{-\lambda T_a} = \mathbb{E}_x(e^{-\lambda \sigma}; T_a < T_b) + \mathbb{E}_x(e^{-\lambda \sigma}; T_b < T_a) \mathbb{E}_b e^{-\lambda T_a}.$$

Interchange the roles of a and b to get a second equation, use Theorem 7.5.7, and solve to get

$$\mathbb{E}_x(e^{-\lambda \sigma}; T_a < T_b) = \frac{\sinh(\sqrt{2\lambda}(b-x))}{\sinh(\sqrt{2\lambda}(b-a))}$$

and

$$\mathbb{E}_x(e^{-\lambda \sigma}; T_b < T_a) = \frac{\sinh(\sqrt{2\lambda}(x-a))}{\sinh(\sqrt{2\lambda}(b-a))}.$$

Knowledge used. [durrett_ch7_strong_markov_property](#);
[durrett_ch7_hitting_time_laplace_exact](#).

Solution. Let

$$A_x = \mathbb{E}_x(e^{-\lambda \sigma}; T_a < T_b), \quad B_x = \mathbb{E}_x(e^{-\lambda \sigma}; T_b < T_a),$$

and put $\alpha = \sqrt{2\lambda}$. On $\{T_a < T_b\}$, we have $T_a = \sigma$. On $\{T_b < T_a\}$, the strong Markov property at σ gives

$$T_a = \sigma + T_a \circ \theta_\sigma,$$

where the restarted Brownian motion begins at b . Hence

$$\mathbb{E}_x e^{-\lambda T_a} = A_x + B_x \mathbb{E}_b e^{-\lambda T_a}.$$

Using the one-sided hitting transform,

$$\mathbb{E}_x e^{-\lambda T_a} = e^{-\alpha(x-a)}, \quad \mathbb{E}_b e^{-\lambda T_a} = e^{-\alpha(b-a)}.$$

Thus

$$e^{-\alpha(x-a)} = A_x + e^{-\alpha(b-a)} B_x.$$

Interchanging a and b similarly gives

$$e^{-\alpha(b-x)} = B_x + e^{-\alpha(b-a)} A_x.$$

Solving the two linear equations yields

$$A_x = \frac{e^{-\alpha(x-a)} - e^{-\alpha(b-a)} e^{-\alpha(b-x)}}{1 - e^{-2\alpha(b-a)}}.$$

Multiplying numerator and denominator by $e^{\alpha(b-a)}$ gives

$$A_x = \frac{e^{\alpha(b-x)} - e^{-\alpha(b-x)}}{e^{\alpha(b-a)} - e^{-\alpha(b-a)}} = \boxed{\frac{\sinh(\alpha(b-x))}{\sinh(\alpha(b-a))}}.$$

The same algebra gives

$$B_x = \boxed{\frac{\sinh(\alpha(x-a))}{\sinh(\alpha(b-a))}}.$$

Exercise 7.5.4

Question. If

$$T = \inf\{t : B_t \notin (a, b)\}, \quad a < 0 < b, \quad a \neq -b,$$

then T and B_T^2 are not independent, so we cannot calculate $\mathbb{E}T^2$ as in the proof of Theorem 7.5.9. Use the Cauchy–Schwarz inequality to estimate $\mathbb{E}(TB_T^2)$ and conclude

$$(i) \mathbb{E}T^2 \leq 4\mathbb{E}B_T^4, \quad (ii) \mathbb{E}B_T^4 \leq 36\mathbb{E}T^2.$$

Knowledge used. [durrett_ch7_heat_equation_martingales](#);
[durrett_ch7_optional_stopping_bounded](#); [durrett_ch7_expected_interval_exit_time](#).

Solution. The polynomial

$$u(t, x) = x^4 - 6tx^2 + 3t^2$$

satisfies $u_t + \frac{1}{2}u_{xx} = 0$, so

$$B_t^4 - 6tB_t^2 + 3t^2$$

is a Brownian martingale. Stopping at $T \wedge n$ and passing to the limit gives

$$\mathbb{E}B_T^4 - 6\mathbb{E}(TB_T^2) + 3\mathbb{E}T^2 = 0.$$

Let

$$X = (\mathbb{E}T^2)^{1/2}, \quad Y = (\mathbb{E}B_T^4)^{1/2}.$$

By Cauchy–Schwarz,

$$\mathbb{E}(TB_T^2) \leq XY.$$

From the martingale identity,

$$3X^2 = 6\mathbb{E}(TB_T^2) - Y^2 \leq 6XY.$$

Hence $X \leq 2Y$, and therefore

$$\boxed{\mathbb{E}T^2 \leq 4\mathbb{E}B_T^4}.$$

The same identity also gives

$$Y^2 = 6\mathbb{E}(TB_T^2) - 3X^2 \leq 6XY.$$

Thus $Y \leq 6X$, and

$$\boxed{\mathbb{E}B_T^4 \leq 36\mathbb{E}T^2}.$$

Exercise 7.5.5

Question. Find a martingale of the form

$$B_t^6 - c_1tB_t^4 + c_2t^2B_t^2 - c_3t^3$$

and use it to compute the third moment of

$$T = \inf\{t : B_t \notin (-a, a)\}.$$

Knowledge used. [durrett_ch7_heat_equation_martingales](#);
[durrett_ch7_symmetric_exit_second_moment](#); [durrett_ch7_optional_stopping_bounded](#).

Solution. Let

$$u(t, x) = x^6 - c_1tx^4 + c_2t^2x^2 - c_3t^3.$$

We choose the constants so that $u_t + \frac{1}{2}u_{xx} = 0$. Now

$$u_t = -c_1x^4 + 2c_2tx^2 - 3c_3t^2,$$

and

$$\frac{1}{2}u_{xx} = 15x^4 - 6c_1tx^2 + c_2t^2.$$

Equating coefficients gives

$$c_1 = 15, \quad c_2 = 45, \quad c_3 = 15.$$

Thus

$$\boxed{B_t^6 - 15tB_t^4 + 45t^2B_t^2 - 15t^3}$$

is a martingale.

For $T = \inf\{t : B_t \notin (-a, a)\}$, we have $B_T = \pm a$. Optional stopping and passage to the limit give

$$0 = \mathbb{E} [B_T^6 - 15TB_T^4 + 45T^2B_T^2 - 15T^3].$$

Therefore

$$0 = a^6 - 15a^4\mathbb{E}T + 45a^2\mathbb{E}T^2 - 15\mathbb{E}T^3.$$

Using $\mathbb{E}T = a^2$ and $\mathbb{E}T^2 = 5a^4/3$,

$$0 = a^6 - 15a^6 + 75a^6 - 15\mathbb{E}T^3 = 61a^6 - 15\mathbb{E}T^3.$$

Hence

$$\boxed{\mathbb{E}T^3 = \frac{61}{15}a^6}.$$

Exercise 7.5.6

Question. Show that

$$(1+t)^{-1/2} \exp\left(\frac{B_t^2}{2(1+t)}\right)$$

is a martingale and use this to conclude that

$$\limsup_{t \rightarrow \infty} \frac{B_t}{\{(1+t) \log(1+t)\}^{1/2}} \leq 1 \quad \text{a.s.}$$

Knowledge used. `durrett_ch7_heat_equation_martingales;`
`durrett_ch7_optional_stopping_bounded;` `durrett_ch7_brownian_martingale.`

Solution. Let

$$u(t, x) = (1+t)^{-1/2} \exp\left(\frac{x^2}{2(1+t)}\right).$$

A direct calculation gives

$$u_x = \frac{x}{1+t} u, \quad u_{xx} = \left(\frac{1}{1+t} + \frac{x^2}{(1+t)^2}\right) u,$$

and

$$u_t = \left(-\frac{1}{2(1+t)} - \frac{x^2}{2(1+t)^2}\right) u.$$

Hence

$$u_t + \frac{1}{2} u_{xx} = 0.$$

By the heat-equation martingale criterion,

$$M_t = (1+t)^{-1/2} \exp\left(\frac{B_t^2}{2(1+t)}\right)$$

is a positive martingale with $M_0 = 1$.

Fix $\varepsilon > 0$ and define

$$c_\varepsilon = \frac{(1+\varepsilon)^2 - 1}{2} > 0.$$

If for some $t \in [e^n, e^{n+1}]$,

$$B_t > (1+\varepsilon)\{(1+t) \log(1+t)\}^{1/2},$$

then

$$M_t \geq (1+t)^{c_\varepsilon} \geq (1+e^n)^{c_\varepsilon}.$$

Doob's maximal inequality for the nonnegative martingale M_t gives

$$\mathbb{P}\left(\sup_{0 \leq s \leq e^{n+1}} M_s \geq (1+e^n)^{c_\varepsilon}\right) \leq (1+e^n)^{-c_\varepsilon}.$$

The right side is summable in n . By Borel–Cantelli, with probability one, only finitely many intervals $[e^n, e^{n+1}]$ contain a time t for which

$$B_t > (1+\varepsilon)\{(1+t) \log(1+t)\}^{1/2}.$$

Therefore

$$\limsup_{t \rightarrow \infty} \frac{B_t}{\{(1+t) \log(1+t)\}^{1/2}} \leq 1 + \varepsilon \quad \text{a.s.}$$

Letting $\varepsilon \downarrow 0$ through rational values yields

$$\boxed{\limsup_{t \rightarrow \infty} \frac{B_t}{\{(1+t) \log(1+t)\}^{1/2}} \leq 1 \quad \text{a.s.}}$$

Exercise 7.6.1

Question. Suppose $f \in C^2$. Show that $f(B_t)$ is a martingale if and only if

$$f(x) = a + bx$$

for constants a, b .

Knowledge used. [durrett_ch7_ito_formula_one_dimensional](#);
[durrett_ch7_stochastic_integral_is_martingale](#); [durrett_ch7_heat_equation_martingales](#).

Solution. If $f(x) = a + bx$, then

$$f(B_t) = a + bB_t,$$

which is a martingale because Brownian motion is a martingale.

Conversely, suppose $f(B_t)$ is a martingale. Ito's formula gives

$$f(B_t) = f(B_0) + \int_0^t f'(B_s) dB_s + \frac{1}{2} \int_0^t f''(B_s) ds.$$

The stochastic integral is a local martingale. Hence the finite variation term

$$\frac{1}{2} \int_0^t f''(B_s) ds$$

must also be a local martingale. A continuous finite variation local martingale is constant, so

$$\int_0^t f''(B_s) ds = 0 \quad \text{for all } t.$$

Taking expectations under $\mathbb{P}_x$ and using the Brownian transition density,

$$0 = \int_0^t \int_{\mathbb{R}} f''(y) p_s(x, y) dy ds.$$

Since f'' is continuous and the heat kernel is strictly positive, this forces $f'' \equiv 0$. Therefore f is affine:

$$\boxed{f(x) = a + bx}.$$

Exercise 7.6.2

Question. Show that

$$B_t^3 - 3tB_t \quad \text{and} \quad B_t^3 - \int_0^t 3B_s ds$$

are martingales.

Knowledge used. [durrett_ch7_ito_formula_one_dimensional](#);
[durrett_ch7_ito_formula_time_space](#); [durrett_ch7_stochastic_integral_is_martingale](#).

Solution. For the first process, let

$$u(t, x) = x^3 - 3tx.$$

Then

$$u_t = -3x, \quad u_x = 3x^2 - 3t, \quad u_{xx} = 6x,$$

so

$$u_t + \frac{1}{2}u_{xx} = -3x + 3x = 0.$$

By the time-space Ito formula,

$$u(t, B_t) = B_t^3 - 3tB_t$$

is a martingale.

For the second process, apply one-dimensional Ito's formula to $f(x) = x^3$:

$$B_t^3 - B_0^3 = \int_0^t 3B_s^2 dB_s + \frac{1}{2} \int_0^t 6B_s ds.$$

Since $B_0 = 0$,

$$B_t^3 - \int_0^t 3B_s ds = \int_0^t 3B_s^2 dB_s,$$

which is a martingale by the stochastic-integral martingale criterion.

Exercise 7.6.3

Question. Let

$$\beta_{2k}(t) = \mathbb{E}_0 B_t^{2k}.$$

Use Ito's formula to relate $\beta_{2k}(t)$ to $\beta_{2k-2}(t)$ and use this relationship to derive a formula for $\beta_{2k}(t)$.

Knowledge used. [durrett_ch7_ito_formula_one_dimensional](#);
[durrett_ch7_stochastic_integral_is_martingale](#).

Solution. Apply Ito's formula to $f(x) = x^{2k}$:

$$B_t^{2k} = \int_0^t 2k B_s^{2k-1} dB_s + \frac{1}{2} \int_0^t 2k(2k-1) B_s^{2k-2} ds.$$

Taking expectations gives

$$\beta_{2k}(t) = k(2k-1) \int_0^t \beta_{2k-2}(s) ds.$$

Equivalently,

$$\beta'_{2k}(t) = k(2k-1)\beta_{2k-2}(t), \quad \beta_0(t) = 1.$$

Induction now gives

$$\beta_{2k}(t) = (2k-1)(2k-3)\cdots 3\cdot 1 t^k.$$

Thus

$$\beta_{2k}(t) = (2k-1)!! t^k = \frac{(2k)!}{2^k k!} t^k.$$

Exercise 7.6.4

Question. Apply Ito's formula to $|B_t|$.

Knowledge used. `durrett_ch7_multidimensional_ito_formula`;
`durrett_ch7_stochastic_integral_is_martingale`.

Solution. For a d -dimensional Brownian motion, let

$$R_t = |B_t|.$$

The function $x \mapsto |x|$ is not C^2 at 0, so one applies Ito's formula first to

$$f_\varepsilon(x) = (|x|^2 + \varepsilon)^{1/2}$$

and then lets $\varepsilon \downarrow 0$. Away from 0,

$$\nabla|x| = \frac{x}{|x|}, \quad \Delta|x| = \frac{d-1}{|x|}.$$

Thus, up to the first hitting time of 0 and, by approximation, for $d \geq 2$ after starting from 0,

$$|B_t| = |B_0| + \sum_{i=1}^d \int_0^t \frac{B_s^i}{|B_s|} dB_s^i + \frac{d-1}{2} \int_0^t \frac{1}{|B_s|} ds.$$

The stochastic integral has quadratic variation t whenever $|B_s| > 0$, since

$$\sum_{i=1}^d \left(\frac{B_s^i}{|B_s|} \right)^2 = 1.$$

So the radial process has the familiar Bessel-form decomposition

$$d|B_t| = dW_t + \frac{d-1}{2|B_t|} dt,$$

where W_t is a one-dimensional Brownian motion generated by the radial martingale part. In dimension $d = 1$, the singularity at 0 contributes the local-time term in Tanaka's formula.

Exercise 7.6.5

Question. Apply Ito's formula to $|B_t|^2$. Use this to conclude that

$$\mathbb{E}_0|B_t|^2 = td.$$

Knowledge used. `durrett_ch7_multidimensional_ito_formula`;
`durrett_ch7_square_martingale`.

Solution. Let $B_t = (B_t^1, \dots, B_t^d)$ and

$$f(x) = |x|^2 = \sum_{i=1}^d x_i^2.$$

Then

$$\frac{\partial f}{\partial x_i} = 2x_i, \quad \frac{\partial^2 f}{\partial x_i^2} = 2.$$

The multidimensional Ito formula gives

$$|B_t|^2 - |B_0|^2 = \sum_{i=1}^d \int_0^t 2B_s^i dB_s^i + \frac{1}{2} \int_0^t \sum_{i=1}^d 2 ds.$$

Since $B_0 = 0$,

$$|B_t|^2 = 2 \sum_{i=1}^d \int_0^t B_s^i dB_s^i + dt.$$

The stochastic integral has mean 0, so

$$\boxed{\mathbb{E}_0 |B_t|^2 = dt}.$$